AI Algorithms: Theoretical Comparison

*Values come from the course slides wherever stated. * = not stated in the slides (s*
Notation: b = branching factor, d = depth of shallowest solution, m = max depth, l =

Table 1: Uninformed Search

Algorithm	Complete
BFS	Yes, if b is finite
UCS	Yes, if step cost >= ε
DFS	No in infinite or looping spaces; yes in finite spaces
DLS	Yes, if l >= d (slide writes l < d, likely a typo)
IDS	Yes
Bidirectional*	Yes (BFS both ways, finite b)*

Table 2: Local Search

Algorithm	Complete
Hill Climbing	No
Simulated Annealing	Not guaranteed in practice
Local Beam Search	No
Genetic Algorithm	No

Table 3: Adversarial Search

Algorithm	Complete
Minimax	Yes (finite game tree)*
Alpha-Beta Pruning	Yes*

MCTS	Yes (in the limit)*

Table 4: CSP

Algorithm	Property
Backtracking Search	Complete; the basic uninformed CSP algorithm
MRV, Most Constraining Variable, LCV	Heuristics that speed up backtracking
Forward Checking	Prunes assignments that guarantee later failure
Arc Consistency (AC-3)	Stronger propagation; detects failure earlier than forward checking
Min-Conflicts	Local search; not complete

slide is an image or only describes the idea); standard textbook value.
= depth limit, C = optimal cost, ε = min step cost.*

Time	Space
O(b^(d+1))	O(b^(d+1)); keeps every node in memory
O(b^ceil(C*/ε))	O(b^ceil(C*/ε))
O(b^m)	O(bm), linear
O(b^l)	O(bl)
O(b^d)	O(bd)
O(b^(d/2))*	O(b^(d/2))*

Optimal	Memory
No; can get stuck on local maxima, shoulders and plateaus	Constant*
Yes in theory, if temperature T is lowered slowly enough	Constant*
No; all k states can end up on the same local maximum	O(k)*
No guarantee (stochastic)	Population size*

Optimal	Time
Yes, against an optimal opponent (perfect play)	O(b^m)*
Yes; same result as minimax	O(b^(m/2)) with perfect move ordering (doubles search depth)

Converges to the optimal move with enough iterations (UCB1)*	Depends on number of iterations*

Details from the slides
DFS with one variable per node, d^n leaves, solves n-queens for n ≈ 25
Combined, they make 1000 queens feasible
Stops when any variable has no legal values
Time O(n^2 d^3)
Solves n-queens in almost constant time, even for n = 10,000,000

Optimal	Key idea
Yes, if step cost is 1	Expand the shallowest node (FIFO queue)
Yes; nodes expanded in increasing cost order	Expand the least-cost node; same as BFS if all costs are equal
No	Expand the deepest node (stack)
No	DFS with a depth cutoff l
Yes, if step cost is 1	Repeated DLS with l = 0, 1, 2, ...
Yes, if step cost is 1*	Search forward from start and backward from goal until they meet

Key idea and limitation
Always move to the best neighbor; fixes: random restart, noise, lookahead
Accepts downhill moves with probability e^(ΔE/T); needs a good annealing schedule
Keeps top k successors of k states; fix: stochastic selection biased toward good states
Selection, crossover and mutation on a population, guided by a fitness function

Space	Key idea
O(bm)*	Generate tree, evaluate terminal states, propagate values upward
O(bm)*	Prune branches when α >= β; move ordering matters

Grows with the tree*	Selection (UCB1), expansion, rollout, backpropagation